

Structure-Enhanced DRL for Optimal Transmission Scheduling

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Abstract—Remote state estimation of large-scale distributed dynamic processes plays an important role in Industry 4.0 applications. In this paper, we focus on the transmission scheduling problem of a remote estimation system. First, we derive some structural properties of the optimal sensor scheduling policy over fading channels. Then, building on these theoretical guidelines, we develop a structure-enhanced deep reinforcement learning (DRL) framework for optimal scheduling of the system to achieve the minimum overall estimation mean-square error (MSE). In particular, we propose a structure-enhanced action selection method, which tends to select actions that obey the policy structure. This explores the action space more effectively and enhances the learning efficiency of DRL agents. Furthermore, we introduce a structure-enhanced loss function to add penalties to actions that do not follow the policy structure. The new loss function guides the DRL to converge to the optimal policy structure quickly. Our numerical experiments illustrate that the proposed structure-enhanced DRL algorithms can save the training time by 50% and reduce the remote estimation MSE by 10% to 25%, when compared to benchmark DRL algorithms. In addition, we show that the derived structural properties exist in a wide range of dynamic scheduling problems that go beyond remote state estimation.

Index Terms—Remote state estimation, deep reinforcement learning, sensor scheduling, threshold structure.

I. INTRODUCTION

WIRELESS networked control systems (WNCSs) consisting of distributed sensors, actuators, controllers, and plants are a key component for Industry 4.0 and have

been widely applied in many areas such as industrial automation, vehicle monitoring systems, building automation, and smart grids [2]. In particular, providing high-quality real-time remote estimation of dynamic system plant states plays an important role in ensuring control performance and stability of WNCSs [3], [4]. For large-scale WNCSs, transmission scheduling of wireless sensors over limited bandwidth needs to be properly designed to guarantee remote estimation performance.

Most existing works on transmission scheduling in wireless communication systems focused on optimizing the quality of service (QoS) performance, such as throughput [5], transmission delay [6], and reliability [7]. In [5], a low-complexity and throughput-optimal scheduling algorithm was proposed for the downlink multi-channel wireless networks. In [6], the authors used a dynamic programming approach to find the delay-optimal scheduling policy for a multi-user-multi-channel system. In [7], cross-layer optimization methods were discussed for delay-aware and reliability-aware scheduler design. However, these conventional transmission scheduling methods are communications-centric and are agnostic to upper-layer applications, i.e., a scheduling policy does not take into account the meaning and importance of each packet. Clearly, in wireless networked control applications, packets carrying different sensing information and control signals play different importance in the overall WNCS performance. Also, the transmission scheduling objective should be the WNCS performance rather than communications QoS.

There are some existing works on task-oriented transmission scheduling of WNCSs [8], [9], [10], [11]. In [8], the optimal scheduling problem of a multi-loop WNCS with limited communication resources was investigated for minimizing the transmission power consumption under a WNCS stability constraint. In [9] and [10], optimal sensor scheduling problems of remote estimation systems were investigated for achieving the best overall estimation MSE. In particular, dynamic decision-making problems were formulated as Markov decision processes (MDPs), which can be solved by classical methods, such as policy and value iterations. The recent work [11] reduces the computation complexity for solving optimal scheduling problems through approximate dynamic programming. In [3], a joint sensor-controller (uplink) and controller-actuator (downlink) transmission scheduling problem was established and solved by the classic MDP framework. Interestingly, the instantaneous remote estimation quality and control performance of the cyber-physical systems

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in [3], [9], and [10] were derived as functions of individual devices' age of information (AoI) states. Thus, AoI is an important metric for characterizing cyber-physical system performance during dynamic scheduling and has attracted lots of attention in the communications society [12]. However, the conventional model-based solutions used in these works are not feasible in large-scale scheduling problems because of the curse of dimensionality caused by high-dimensional state and action spaces.¹

In recent years, deep reinforcement learning (DRL) has been developed to deal with large MDPs by using deep neural networks as function approximators [13], [14]. Some works [15], [16], [17], [18] have used the deep Q-network (DQN), a simple DRL method, to solve multi-sensor-multi-channel scheduling problems in different remote estimation scenarios. In particular, sensor scheduling problems for systems with 6 sensors have been solved effectively, providing significant performance gains over heuristic methods in terms of estimation quality. The more recent work [19] has introduced DRL algorithms with an actor-critic structure to solve scheduling problems at a much larger scale (that cannot be handled by the DQN).² However, existing works merely use the general DRL frameworks to solve specific scheduling problems without questioning what features distinguish sensor scheduling problems from other MDPs. Also, we note that the drawback of general DRL is that it often cannot perform policy exploration effectively for specific tasks [21], which can lead to getting stuck in local minima or even total failure. Thus, the existing DRL-based solutions could perform far from optimal. We propose a novel question: *is it possible to investigate the structure of the optimal policy of a specific scheduling problem first and then utilize the policy structure for tailored DRL algorithm design to find the optimal policy effectively?*

A key feature that we will exploit in our current work is that optimal transmission scheduling policies of remote estimation systems often have threshold structures [10], [22], [23], which means that there exist switching boundaries dividing the state space into multiple regions for different scheduling actions. In other words, an optimal policy has a structure where the action only changes at the switching boundaries of the state space. In particular, [22] focuses on an energy-constrained single-sensor-single-channel system and proves that the optimal policy has a threshold in terms of the sensor's AoI, determining whether the sensor will be scheduled or not. In [23], the authors considered a multi-sensor-multi-channel system, where all channels are static, and each sensor has a constant packet-drop probability at all frequency channels. This work also showed that the optimal scheduling policy has a multi-dimensional threshold structure in terms of all sensor AoI. As an extension, the work in [10] (a two-sensor system), assumed that different sensors could have different numbers

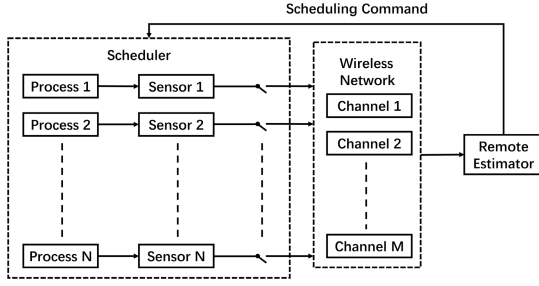
of packets for carrying each measurement. The demonstrated threshold property of the optimal policy is related to the sensor AoI and the remaining packet numbers of each sensor. There are two limitations of the channel models adopted in the above works: 1) fading channel models are commonly adopted in practice, where channel states are time-varying, and 2) wireless propagation via different frequency bandwidths has different properties, leading to different channel qualities. Not limited to remote estimation systems, in [24], the threshold structure of an optimal sensor scheduling policy has also been identified for minimizing the average sum AoI. However, this work only considered a single-channel system, and the transmission success or failure was determined before a scheduling action. Therefore, the theoretical works [10], [22], [23], [24] only derived the structural properties of optimal policies under some ideal assumptions. It is an open challenge to prove the existence of structural results of the optimal scheduling policy of a general multi-sensor-multi-channel system over practical fading channels. Furthermore, there is no existing work in the open literature utilizing the structural properties to guide DRL algorithms for effectively solving optimal scheduling problems.

In this paper, we consider the optimal sensor scheduling problem of a general N -sensor- M -channel remote estimation system over fading channels. The main contributions of this work are summarized as follows.

- We prove that the optimal sensor scheduling policy has a threshold structure in terms of both the AoI states of all sensors and the corresponding channel states, where the channel states of different sensors at different frequencies are different. To the best of our knowledge, this is the first structural result of optimal scheduling policies over fading channels in the literature. In addition, we show that such a structural property exists in a wide range of dynamic scheduling/resource allocation problems, not limited to remote state estimations. Interestingly, we also give a counterexample to show when such a property does not exist.
- We first formulate the sensor scheduling problem into an MDP, and then develop novel structure-enhanced DRL algorithms for solving the problem, building on the derived threshold properties of the optimal policy. In particular, we design a **structure-enhanced action selection method**, which tends to select actions that obey the threshold structure. Such an action selection method can explore the action space more effectively and enhance the learning efficiency of DRL agents. Furthermore, we introduce a **structure-enhanced loss function** to add penalties to actions that do not follow the threshold structure. The new loss function guides the DRL to converge to the optimal policy structure quickly. We apply the proposed action selection method and the novel loss function to redesign the most commonly adopted DRL frameworks for scheduling, i.e., DQN and deep deterministic policy gradient (DDPG), referred to as the **structure-enhanced DQN and DDPG algorithms**.
- Our extensive numerical results illustrate that the proposed structure-enhanced DRL algorithms can save the

¹In [8], [9], [10], only the optimal scheduling of two-sensor systems has been solved effectively by the proposed methods.

²The works [15], [16], [17], [18] adopted the orthogonal multiple-access (OMA) scheme in the remote estimation system, while [19] considered non-orthogonal multiple-access (NOMA) schemes. For OMA, each channel can only be allocated to a single sensor; for NOMA, each frequency channel can be assigned to multiple sensors simultaneously, and the packets from different sensors are detected by processing the received superimposed signals [20].

Fig. 1. Remote state estimation system with N processes and M channels.

training time by 50% while reducing the remote estimation MSE by 10% to 25% compared with benchmark DRL algorithms. Importantly, the structure-enhanced DRL algorithms can converge and perform well under some system settings that cannot be solved by any of the benchmark DRL algorithms.

Outline: The system model of the remote state estimation system is described in Section II. The sensor scheduling problem formulation and the introduction of the structural properties are presented in Section III. The proofs of structural properties of the optimal scheduling policy are given in Section IV. The structure-enhanced DRL algorithms for the formulated problem are presented in Section V. The numerical results are shown and analyzed in Section VI, followed by conclusions in Section VII.

Notations: Matrices and vectors are denoted by capital and lowercase upright bold letters, e.g., $\mathbf{A}$ and $\mathbf{a}$, respectively. $\mathbb{E}[\cdot]$ is the expectation operator. $(\cdot)^\top$ is the vector or matrix transpose operator. $(\cdot)^{-1}$ is the matrix inverse operator. $\text{Tr}(\cdot)$ is the matrix trace operator. $\text{Pr}(\cdot)$ and $\text{Pr}(\cdot|\cdot)$ denote the probability and conditional probability, respectively. $\mathbb{N}$ and $\mathbb{R}$ denote the sets of positive integers and real numbers, respectively.

II. SYSTEM MODEL

We consider a remote estimation system with N dynamic processes, each measured by a sensor, which pre-processes the raw measurements and sends its state estimates to a remote estimator through one of M wireless channels, as illustrated in Fig. 1.

A. Dynamic Process Model and Local State Estimation

Each dynamic process n is modeled as a discrete-time linear time-invariant (LTI) system as [15], [25], and [26]

$$\begin{aligned} \mathbf{x}_{n,t+1} &= \mathbf{A}_n \mathbf{x}_{n,t} + \mathbf{w}_{n,t}, \\ \mathbf{y}_{n,t} &= \mathbf{C}_n \mathbf{x}_{n,t} + \mathbf{v}_{n,t}, n \in \{1, \dots, N\}, t \in \{1, \dots\}, \end{aligned} \quad (1)$$

where $\mathbf{x}_{n,t} \in \mathbb{R}^{l_n}$ is process n 's state at time t , and $\mathbf{y}_{n,t} \in \mathbb{R}^{e_n}$ is the state measurement of the sensor n , $\mathbf{A}_n \in \mathbb{R}^{l_n \times l_n}$ and $\mathbf{C}_n \in \mathbb{R}^{e_n \times l_n}$ are the system matrix and the measurement matrix, respectively, $\mathbf{w}_{n,t} \in \mathbb{R}^{l_n}$ and $\mathbf{v}_{n,t} \in \mathbb{R}^{e_n}$ are the process disturbance and the measurement noise modeled as independent and identically distributed (i.i.d) zero-mean Gaussian random vectors $\mathcal{N}(\mathbf{0}, \mathbf{W}_n)$ and $\mathcal{N}(\mathbf{0}, \mathbf{V}_n)$, respectively. We assume that the spectral radius of $\mathbf{A}_n, \forall n$, is greater

than one, which means that the dynamic processes are non-stationary,³ making the remote estimation problem more interesting (see [25] and references therein).

Due to the presence of noise in (1), each sensor n executes a classic Kalman filter to pre-process the raw measurement and generate state estimate $\mathbf{x}_{n,t}^s$ at each time t as [25]

$$\mathbf{x}_{n,t|t-1}^s = \mathbf{A}_n \mathbf{x}_{n,t-1}^s \quad (2a)$$

$$\mathbf{P}_{n,t|t-1}^s = \mathbf{A}_n \mathbf{P}_{n,t-1}^s \mathbf{A}_n^\top + \mathbf{W}_n \quad (2b)$$

$$\mathbf{K}_{n,t} = \mathbf{P}_{n,t|t-1}^s \mathbf{C}_n^\top (\mathbf{C}_n \mathbf{P}_{n,t|t-1}^s \mathbf{C}_n^\top + \mathbf{V}_n)^{-1} \quad (2c)$$

$$\mathbf{x}_{n,t}^s = \mathbf{x}_{n,t|t-1}^s + \mathbf{K}_{n,t} (\mathbf{y}_{n,t} - \mathbf{C}_n \mathbf{x}_{n,t|t-1}^s) \quad (2d)$$

$$\mathbf{P}_{n,t}^s = (\mathbf{I}_n - \mathbf{K}_{n,t} \mathbf{C}_n) \mathbf{P}_{n,t|t-1}^s \quad (2e)$$

where $\mathbf{x}_{n,t|t-1}^s$ and $\mathbf{P}_{n,t|t-1}^s$ are the prior state estimate and the corresponding estimation error covariance of sensor n , respectively, $\mathbf{x}_{n,t}^s$ and $\mathbf{P}_{n,t}^s$ are the posterior state estimate and the corresponding estimation error covariance of sensor n at time t , respectively.

In particular, sensor n sends the estimate $\mathbf{x}_{n,t}^s$ to the remote estimator (not $\mathbf{y}_{n,t}$) as a packet, once scheduled, and the local state estimation error covariance matrix is defined as

$$\mathbf{P}_{n,t}^s \triangleq \mathbb{E}[(\mathbf{x}_{n,t}^s - \mathbf{x}_{n,t})(\mathbf{x}_{n,t}^s - \mathbf{x}_{n,t})^\top]. \quad (3)$$

$\mathbf{K}_{n,t}$ is the Kalman gain of sensor n , and $\mathbf{I}_n$ is an identity matrix. Note that (2a) and (2b) present the prediction steps while (2c), (2d), and (2e) are the updating steps. We note that local Kalman filters are commonly assumed to operate in the steady state mode in the literature (see [25] and references therein⁴). We thus assume that the error covariance matrix has converged to a constant, i.e., $\mathbf{P}_{n,t}^s = \bar{\mathbf{P}}_n, \forall t, n$.

B. Wireless Communications and Remote State Estimation

There are only M wireless channels (e.g., subcarriers) for the N sensors' transmissions, where $M \leq N$. We consider independent and identically distributed (i.i.d.) block fading channels, where the channel quality is fixed during each packet transmission and varies packet by packet, independently. Let the $N \times M$ matrix $\mathbf{H}_t$ denote the channel state of the system at time t , where the element in the n th row and m th column, say $h_{n,m,t} \in \mathcal{H} \triangleq \{1, 2, \dots, \bar{h}\}$, represents the channel state between sensor n and the remote estimator at channel m . In particular, there are $\bar{h}$ quantized channel states in total.⁵ The distribution of $h_{n,m,t}$ is given as

$$\text{Pr}(h_{n,m,t} = i) = q_i^{(n,m)}, \forall t, \quad (4)$$

where $\sum_{i=1}^{\bar{h}} q_i^{(n,m)} = 1, \forall n, m$. The instantaneous full channel state $\mathbf{H}_t$ is available at the remote estimator based on standard channel estimation methods [27]. The estimated channel state will be used for effectively generating the scheduling command, see e.g., [28], [29], [30], [31].

³When the spectral radius of $\mathbf{A}_n$ is greater than 1, the plant state, $\mathbf{x}_{n,t}$, grows exponentially fast and hence the distribution of $\mathbf{x}_{n,t}$ changes over time.

⁴The n th local Kalman filter converges to steady state if $(\mathbf{A}_n, \mathbf{C}_n)$ is observable and $(\mathbf{A}_n, \sqrt{\mathbf{W}_n})$ is controllable.

⁵We note that the discrete channel states are quantized from continuous fading channels in practice, such as Rayleigh, Rician, and Nakagami channels; the channel state distributions can be obtained by discretizing the corresponding continuous distribution.

The packet drop probability at channel state $i' \in \mathcal{H}$ is denoted as $\tilde{p}_{i'}$. Without loss of generality, we assume that $\tilde{p}_1 \geq \tilde{p}_2 \geq \dots \geq \tilde{p}_h$. We also define the packet success rate for the channel state $h_{n,m,t}$ as $p_{n,m,t} \in \{p_1, \dots, p_h\}$, where $p_{i'} \triangleq 1 - \tilde{p}_{i'}$.

Due to the limited communication channels, only M out of N sensors can be scheduled at each time step. Let $a_{n,t} \in \{0, 1, 2, \dots, M\}$ represent the channel allocation for sensor n at time t , where

$$a_{n,t} = \begin{cases} 0, & \text{if sensor } n \text{ is not scheduled} \\ m, & \text{if sensor } n \text{ is scheduled to channel } m. \end{cases}$$

In particular, we assume that each sensor can be scheduled to at most one channel and that each channel is assigned to one sensor [19]. Then, the constraints on $a_{n,t}$ are given as

$$\sum_{m=1}^M \mathbb{1}(a_{n,t} = m) \leq 1, \quad \sum_{n=1}^N \mathbb{1}(a_{n,t} = m) = 1, \quad (5)$$

where $\mathbb{1}(\cdot)$ is the indicator function.

Considering schedule actions and packet dropouts, sensor n 's estimate may not be received by the remote estimator in every time slot. We define the packet reception indicator as

$$\eta_{n,t} = \begin{cases} 1, & \text{if sensor } n\text{'s packet is received at time } t \\ 0, & \text{otherwise.} \end{cases}$$

Considering the randomness of $\eta_{n,t}$ and assuming that the remote estimator performs state estimation at the beginning of each time slot, the remote state estimate that minimizes the estimation MSE follows the stochastic recursion:

$$\hat{\mathbf{x}}_{n,t+1} = \begin{cases} \mathbf{A}_n \mathbf{x}_{n,t}^s, & \text{if } \eta_{n,t} = 1 \\ \mathbf{A}_n \hat{\mathbf{x}}_{n,t}, & \text{otherwise,} \end{cases} \quad (6)$$

where $\mathbf{A}_n$ is the system matrix of process n defined in (1). If sensor n 's packet is not received, then the remote estimator propagates its estimate in the previous time slot to estimate the current state. From (1) and (6), we derive the estimation error covariance as

$$\begin{aligned} \mathbf{P}_{n,t} &\triangleq \mathbb{E} \left[(\hat{\mathbf{x}}_{n,t} - \mathbf{x}_{n,t}) (\hat{\mathbf{x}}_{n,t} - \mathbf{x}_{n,t})^\top \right] \\ &= \begin{cases} \mathbf{A}_n \bar{\mathbf{P}}_n \mathbf{A}_n^\top + \mathbf{W}_n, & \text{if } \eta_{n,t} = 1 \\ \mathbf{A}_n \mathbf{P}_{n,t} \mathbf{A}_n^\top + \mathbf{W}_n, & \text{otherwise,} \end{cases} \end{aligned} \quad (7)$$

where $\bar{\mathbf{P}}_n$ is the local estimation error covariance of sensor n defined under (3).

Let $\tau_{n,t} \in \{1, 2, \dots\}$ denote the age-of-information (AoI) of sensor n at time t , which measures the amount of time elapsed since the latest sensor packet was successfully received. Then, we have

$$\tau_{n,t+1} = \begin{cases} 1, & \text{if } \eta_{n,t} = 1 \\ \tau_{n,t} + 1, & \text{otherwise.} \end{cases} \quad (8)$$

If a sensor is frequently scheduled at good channels, then the corresponding average AoI is small. However, due to the scheduling constraint (5), this is often not possible.

From (7) and (8), the error covariances can be written in terms of the AoI as

$$\mathbf{P}_{n,t} = f_n^{\tau_{n,t}}(\bar{\mathbf{P}}_n), \quad (9)$$

where $f_n(\mathbf{X}) = \mathbf{A}_n \mathbf{X} \mathbf{A}_n^\top + \mathbf{W}_n$ and $f_n^{\tau+1}(\cdot) = f_n(f_n^\tau(\cdot))$. It has been proved that the estimation MSE, i.e., $\text{Tr}(\mathbf{P}_{n,t})$, *monotonically increases with the AoI state $\tau_{n,t}$* [25].

III. PROBLEM FORMULATION AND THRESHOLD STRUCTURE

In this paper, we aim to find a dynamic scheduling policy $\pi(\cdot)$ that uses the AoI states of all sensors, as well as the channel states to minimize the expected total discounted estimation MSE of all N processes over the infinite time horizon.

Problem 1:

$$\max_{\pi} \lim_{T \rightarrow \infty} \mathbb{E} \left[\sum_{t=1}^T \sum_{n=1}^N -\gamma^t \text{Tr}(\mathbf{P}_{n,t}) \right],$$

where $\gamma \in (0, 1)$ is a discount factor.

Problem 1 is a Markovian sequential decision-making problem. This is because the instantaneous estimation MSE, $\mathbf{P}_{n,t}$, only depends on the AoI state $\tau_{n,t}$ in (9), which is Markovian (8), and the channel states are i.i.d.. Therefore, we formulate Problem 1 as an MDP.⁶

A. MDP Formulation

Given the observable state $\mathbf{s}_t$, including both the channel state and the AoI state, the scheduler generates an action $\mathbf{a}_t$ based on a policy π which is a state-action mapping, i.e., $\mathbf{a}_t = \pi(\mathbf{s}_t)$. An immediate reward is generated to measure the overall estimation quality at the current time step, i.e., the negative sum of estimation MSE, i.e., $\sum_{n=1}^N -\text{Tr}(\mathbf{P}_{n,t})$, which is determined by $\mathbf{s}_t$. Due to the Markovian property of the decision-making process, the probability of state transitioning from one to the next only relies on the current state and action, which can be written as $\Pr(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$. Under the state transition rule, the overall target of the MDP problem is to optimize the policy $\pi(\cdot)$ for maximizing the expected total discounted reward over the infinite time horizon. The formal definition of the MDP are given below:

1) The state of the MDP, taking into account both the AoI and channel states, is defined as $\mathbf{s}_t \triangleq (\boldsymbol{\tau}_t, \mathbf{H}_t) \in \mathcal{S} \triangleq \mathbb{N}^N \times \mathcal{H}^{N \times M}$, where $\boldsymbol{\tau}_t = (\tau_{1,t}, \tau_{2,t}, \dots, \tau_{N,t}) \in \mathbb{N}^N$ is the AoI state vector.

2) The overall schedule action of the N sensors is defined as $\mathbf{a}_t = (a_{1,t}, a_{2,t}, \dots, a_{N,t}) \in \mathcal{A} \triangleq \{0, 1, 2, \dots, M\}^N$ under the constraint (5). There are $N!/(N-M)!$ actions of the N -choose- M problem in total.

3) The transition probability $\Pr(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$ is the probability of the next state $\mathbf{s}_{t+1}$ given the current state $\mathbf{s}_t$ and

⁶Not all MDPs for transmission scheduling of remote estimation systems have a feasible solution. However, once the remote estimation stability condition in terms of the dynamic process parameters and the channel statistics is satisfied, the MDP has a solution. We assume that the remote estimation stability condition of our system is satisfied, and only focus on the optimal solution of the MDP in the rest of the paper. The detailed stability condition can be found in our previous work [19].

the action $\mathbf{a}_t$. Since the state transition is independent of the time index given the action $\mathbf{a}$ and the state $\mathbf{s}$, we drop the subscript t here and use $\mathbf{s}$ and $\mathbf{s}^+$ to represent the current and the next states, respectively. Due to the i.i.d. fading channel states, we have

$$\Pr(\mathbf{s}^+|\mathbf{s}, \mathbf{a}) = \Pr(\boldsymbol{\tau}^+|\boldsymbol{\tau}, \mathbf{H}, \mathbf{a}) \Pr(\mathbf{H}^+), \quad (10)$$

where $\Pr(\mathbf{H}^+)$ can be obtained from (4), and $\Pr(\boldsymbol{\tau}^+|\boldsymbol{\tau}, \mathbf{H}, \mathbf{a}) = \prod_{n=1}^N \Pr(\tau_n^+|\tau_n, \mathbf{H}, a_n)$ and

$$\Pr(\tau_n^+|\tau_n, \mathbf{H}, a_n) = \begin{cases} p_{n,m}, & \text{if } \tau_n^+ = 1, a_n = m \\ 1 - p_{n,m}, & \text{if } \tau_n^+ = \tau_n + 1, a_n = m \\ 1, & \text{if } \tau_n^+ = \tau_n + 1, a_n = 0 \\ 0, & \text{otherwise,} \end{cases}$$

which is derived based on (4) and (8).

4) The immediate reward at time t is defined as the negative sum estimation MSE, i.e., $\sum_{n=1}^N -\text{Tr}(\mathbf{P}_{n,t})$, to measure the overall remote estimation quality. Since $\mathbf{P}_{n,t}$ defined in (9) is a function of the AoI state $\tau_{n,t}$, the reward is denoted as $r(\mathbf{s}_t)$, monotonically decreasing with each AoI state.

Note that the formulated MDP problem cannot be solved by conventional algorithms such as value and policy iterations due to the high computation complexity even for small-scale systems (e.g., $N = 6$ and $M = 3$) [32].

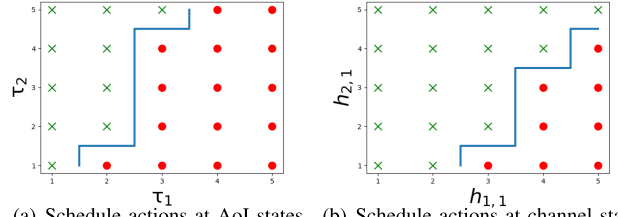
B. Threshold Structure of the Optimal MDP Solution

Unlike the existing works [10], [23], which only considered oversimplified systems over static channels, we aim to derive the structural property of the optimal policy of the general multi-sensor-multi-channel system over fading channels as below.

Definition 1 (Channel-State Threshold Policy): For a channel-state threshold scheduling policy, if channel m is assigned to sensor n at the state $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, then for state $\mathbf{s}' = (\boldsymbol{\tau}, \mathbf{H}'_{n,m})$, where $\mathbf{H}'_{n,m}$ is identical to $\mathbf{H}$ except the sensor- n -channel- m state with $h'_{n,m} > h_{n,m}$, then channel m is still assigned to sensor n .

Definition 2 (AoI-State Threshold Policy): For an AoI-state threshold scheduling policy, if channel m is assigned to sensor n at the state $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, then for state $\mathbf{s}' = (\boldsymbol{\tau}'_{(n)}, \mathbf{H})$, where $\boldsymbol{\tau}'_{(n)}$ is identical to $\boldsymbol{\tau}$ except sensor n 's AoI with $\tau'_n \geq \tau_n$, then either channel m or a better channel is assigned to sensor n .

Definition 1 states that sensor n is scheduled at channel m at a certain state, if the channel quality of $h_{n,m}$ improves while the AoI and the other channel states are the same, then the threshold policy still assigns channel m to sensor n . For Definition 2, if the AoI state of sensor n is increased while the other states remain the same, the policy must schedule sensor n to a channel that is no worse than the previous one. To illustrate that an optimal policy may have a threshold structure, we find the optimal policy of a two-sensor-single-channel system by solving the MDP with the conventional value iteration algorithm, which is illustrated in Fig. 2. We see that action-switching curves exist in both AoI and channel state spaces, and the properties in Definitions 1 and 2 are observed. Inspired by the above result, in the



(a) Schedule actions at AoI states (b) Schedule actions at channel states
Fig. 2. Structure of the optimal scheduling policy with $N = 2$ and $M = 1$, where $\bullet$ and $\times$ represent the schedule of sensor 1 and 2, respectively.

following, we will prove that the optimal policy has the structural properties in Definitions 1 and 2.⁷

IV. PROOF OF THE THRESHOLD STRUCTURE OF OPTIMAL POLICIES

We derive the structural properties of the optimal scheduling policy by using value iteration concepts, which require the definition of the optimal value function, $V^*(\mathbf{s}_t) : \mathcal{S} \rightarrow \mathbb{R}$, and the state-action value function, $Q(\mathbf{s}_t, \mathbf{a}_t) : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ as below.

Given the current state $\mathbf{s}_t = (\boldsymbol{\tau}_t, \mathbf{H}_t)$, the optimal value function is the maximum expected discounted sum of the future reward, i.e., achieved by the optimal policy $\pi^*(\cdot)$:

$$V^*(\mathbf{s}_t) = \max_{\pi} \mathbb{E} \left[\sum_{t'=t}^{\infty} \gamma^{t'-t} r(\mathbf{s}_{t'}) \right].$$

The optimal value function satisfies the Bellman equation:

$$V^*(\mathbf{s}_t) = r(\mathbf{s}_t) + \gamma \max_{\mathbf{a}_t \in \mathcal{A}} \left[\sum_{\mathbf{s}_{t+1}} \Pr(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) V^*(\mathbf{s}_{t+1}) \right], \quad (11)$$

where the optimal action $\mathbf{a}^*$ is obtained by the optimal policy $\pi^*(\cdot)$, i.e.,

$$\mathbf{a}_t^* \triangleq \pi^*(\mathbf{s}_t) = \arg \max_{\mathbf{a}_t \in \mathcal{A}} \left[\sum_{\mathbf{s}_{t+1}} \Pr(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) V^*(\mathbf{s}_{t+1}) \right].$$

Given the current state-action pair $\mathbf{s}_t$ and $\mathbf{a}_t$, the state-action value function, which is also called the Q-value function, measures the expected discounted sum of the future cost under the optimal policy $\pi^*(\cdot)$ as

$$Q(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t) + \gamma \sum_{\mathbf{s}_{t+1}} \Pr(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) V^*(\mathbf{s}_{t+1}). \quad (12)$$

From (11) and (12), it directly follows that the optimal value function and the Q-value function satisfy:

$$V^*(\mathbf{s}_t) = Q(\mathbf{s}_t, \mathbf{a}_t^*) \geq Q(\mathbf{s}_t, \mathbf{a}_t).$$

For notation simplicity, we use $\mathbf{a}, \mathbf{s}, \mathbf{s}^+$ to represent $\mathbf{a}_t, \mathbf{s}_t, \mathbf{s}_{t+1}$ in the following.

⁷We note that in [10] and [23], only threshold structural results in terms of the AoI states are proved over static channels. Moreover, the proofs have some limitations: 1) In [10], the authors proved the threshold structure for a multi-dimensional state and action scenario based on Theorem 8.11.3 in [32], which, however, can only be used for single-dimensional scenarios. 2) In [23], the proof of the optimal policy's structural property only considered a part of the action space; however, the full action space must be examined to show the optimality.

We prove the channel-state and the AoI-state threshold properties of the optimal scheduling policy based on the classical value iteration algorithm, as it can achieve the optimal solution [32], [33]. In the value iteration, the initial value function and its $\tilde{t}$ -th iteration are $V^0(\mathbf{s}) \in \mathcal{V}$ and $V^{\tilde{t}}(\mathbf{s}) \in \mathcal{V}$, respectively, where $\mathcal{V}$ is the set of any measurable function, i.e., $\mathcal{V} : \mathcal{S} \rightarrow \mathbb{R}$. At the $\tilde{t}$ -th iteration, we have $V^{\tilde{t}+1} = \mathcal{B}[V^{\tilde{t}}]$, where $\mathcal{B}[\cdot] : \mathcal{V} \rightarrow \mathcal{V}$ is the Bellman operator:

$$\mathcal{B}[V^{\tilde{t}}](\mathbf{s}) = r(\mathbf{s}) + \gamma \max_{\mathbf{a} \in \mathcal{A}} \left[\sum_{\mathbf{s}^+} \Pr(\mathbf{s}^+ | \mathbf{s}, \mathbf{a}) V^{\tilde{t}}(\mathbf{s}^+) \right].$$

We next elucidate the convergence and the optimality of value iterations.

Lemma 1 ([32], [33]): *If the optimal policy exists, then the operator $\mathcal{B}$ has a unique fixed point $V^* \in \mathcal{V}$ and for all $V^0 \in \mathcal{V}$, the sequence $\{V^{\tilde{t}}\}$ defined by $V^{\tilde{t}+1} = \mathcal{B}[V^{\tilde{t}}]$ converges in norm to V^* , i.e.*

$$\lim_{\tilde{t} \rightarrow \infty} V^{\tilde{t}} = V^*.$$

Before proceeding further, we derive the following technical lemma about the monotonicity of the optimal value function.

Lemma 2 (Monotonicity): *Consider states $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$ and $\mathbf{s}' = (\boldsymbol{\tau}'_{(i)}, \mathbf{H})$, where $\boldsymbol{\tau}'_{(i)} = (\tau_1, \dots, \tau'_i, \dots, \tau_N)$ and $\tau'_i \geq \tau_i$. The following holds*

$$V^*(\mathbf{s}') \leq V^*(\mathbf{s}).$$

Proof: See Appendix A in [34], due to space limitations. $\square$

Lemma 2 will assist in the comparison of value functions with different AoI states in the proof of the channel-state threshold and the AoI-state threshold properties.

A. Channel-State Threshold Property

In this part, we completely prove that the optimal policy of a general multi-sensor-multi-channel system is of the channel-state threshold property.

Theorem 1: *The optimal policy $\pi^*(\cdot)$ of the multi-sensor-multi-channel system has the channel-state threshold property in Definition 1.*

Proof: From the Q-value definition in (12), the theorem can be translated as: if for state $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, the inequality $Q(\mathbf{s}, \mathbf{a}^*) \geq Q(\mathbf{s}, \mathbf{a}), \forall \mathbf{a} \in \mathcal{A}$ exists, where $a_i^* = m$, then for state $\mathbf{s}' = (\boldsymbol{\tau}, \mathbf{H}'_{i,m})$, where $\mathbf{H}'_{i,m}$ and $\mathbf{H}$ are identical except the element $h'_{i,m} > h_{i,m}$, the following inequality holds $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a})$, where $\mathbf{a}^*$ is the optimal action at the state $\mathbf{s}'$ with $a_i^* = m$. Since $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a}^*)$, we only need to prove $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a}')$, where $a'_i \neq m$. To prove $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a}')$, we will prove $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}, \mathbf{a}^*)$ and then $Q(\mathbf{s}, \mathbf{a}') = Q(\mathbf{s}', \mathbf{a}')$. In the following, we use $\mathbf{H}'$ to represent $\mathbf{H}'_{i,m}$ for the notation simplicity.

First, from (4) and (8), we have

$$\begin{aligned} \Pr(\boldsymbol{\tau}^+ | \boldsymbol{\tau}, \mathbf{H}, \mathbf{a}) &= \prod_{n=1}^N \Pr(\tau_n^+ | \tau_n, \mathbf{h}_n, a_n) \\ &= \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a_i) \Pr(\boldsymbol{\tau}_{\setminus i}^+ | \boldsymbol{\tau}_{\setminus i}, \mathbf{H}_{\setminus i}, \mathbf{a}_{\setminus i}), \end{aligned} \quad (13)$$

where $\boldsymbol{\tau}_{\setminus i} = (\tau_1, \dots, \tau_{i-1}, \tau_{i+1}, \dots, \tau_N)$ and $\mathbf{a}_{\setminus i} = (a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_N)$ represent the AoI states and the actions of all sensors except sensor i , respectively, and $\mathbf{h}_i = (h_{i,1}, h_{i,2}, \dots, h_{i,M})$ is the vector channel states between sensor i and the remote estimator and $\mathbf{H}_{\setminus i} = (\mathbf{h}_1, \dots, \mathbf{h}_{i-1}, \mathbf{h}_{i+1}, \dots, \mathbf{h}_N)$. By using (10) and (13), it can be derived that

$$\begin{aligned} Q(\mathbf{s}, \mathbf{a}) &= r(\mathbf{s}) + \gamma \sum_{\mathbf{s}^+} \Pr(\mathbf{s}^+ | \mathbf{s}, \mathbf{H}, \mathbf{a}) V^*(\mathbf{s}^+) \\ &= r(\mathbf{s}) + \gamma \sum_{\mathbf{H}^+} \sum_{\boldsymbol{\tau}^+} \Pr(\mathbf{H}^+) \Pr(\boldsymbol{\tau}^+ | \boldsymbol{\tau}, \mathbf{H}, \mathbf{a}) V^*(\mathbf{s}^+) \\ &= r(\mathbf{s}) + \gamma \sum_{\mathbf{H}^+} \sum_{\boldsymbol{\tau}_{\setminus i}^+} \sum_{\tau_i^+} \Pr(\mathbf{H}^+) \Pr(\boldsymbol{\tau}_{\setminus i}^+ | \boldsymbol{\tau}_{\setminus i}, \mathbf{H}_{\setminus i}, \mathbf{a}_{\setminus i}) \\ &\quad \times \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a_i) V^*(\mathbf{s}^+). \end{aligned} \quad (14)$$

Based on (14), we have

$$\begin{aligned} Q(\mathbf{s}', \mathbf{a}^*) &= r(\mathbf{s}') + \gamma \sum_{\mathbf{H}'^+} \sum_{\boldsymbol{\tau}_{\setminus i}^+} \sum_{\tau_i^+} \Pr(\mathbf{H}'^+) \Pr(\boldsymbol{\tau}_{\setminus i}^+ | \boldsymbol{\tau}_{\setminus i}, \mathbf{H}_{\setminus i}, \mathbf{a}_{\setminus i}^*) \\ &\quad \times \Pr(\tau_i^+ | \tau_i, \mathbf{h}'_i, a_i^*) V^*(\mathbf{s}'^+) \\ &\geq r(\mathbf{s}) + \gamma \sum_{\mathbf{H}^+} \sum_{\boldsymbol{\tau}_{\setminus i}^+} \sum_{\tau_i^+} \Pr(\mathbf{H}^+) \Pr(\boldsymbol{\tau}_{\setminus i}^+ | \boldsymbol{\tau}_{\setminus i}, \mathbf{H}_{\setminus i}, \mathbf{a}_{\setminus i}^*) \\ &\quad \times \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a_i^*) V^*(\mathbf{s}^+) \\ &= Q(\mathbf{s}, \mathbf{a}^*), \end{aligned}$$

where the inequality is derived by replacing the parameter $\mathbf{H}'^+$ with $\mathbf{H}^+$, and then using $r(\mathbf{s}') = r(\mathbf{s})$, $h'_{i,m} > h_{i,m}$, $a_i^* = m$, and the following inequality

$$\begin{aligned} p'_{i,m} V^*(1, \boldsymbol{\tau}_{\setminus i}^+, \mathbf{H}^+) &+ (1 - p'_{i,m}) V^*(\tau_i + 1, \boldsymbol{\tau}_{\setminus i}^+, \mathbf{H}^+) \\ &\geq p_{i,m} V^*(1, \boldsymbol{\tau}_{\setminus i}^+, \mathbf{H}^+) + (1 - p_{i,m}) V^*(\tau_i + 1, \boldsymbol{\tau}_{\setminus i}^+, \mathbf{H}^+) \end{aligned}$$

achieved by $p'_{i,m} \geq p_{i,m}$ and $V^*(1, \boldsymbol{\tau}_{\setminus i}^+, \mathbf{H}^+) \geq V^*(\tau_i + 1, \boldsymbol{\tau}_{\setminus i}^+, \mathbf{H}^+)$ from Lemma 2.

Second, we derive that

$$\begin{aligned} Q(\mathbf{s}, \mathbf{a}) &= r(\mathbf{s}) + \gamma \sum_{\mathbf{H}^+} \sum_{\boldsymbol{\tau}_{\setminus i}^+} \sum_{\tau_i^+} \Pr(\mathbf{H}^+) \Pr(\boldsymbol{\tau}_{\setminus i}^+ | \boldsymbol{\tau}_{\setminus i}, \mathbf{H}_{\setminus i}, \mathbf{a}'_{\setminus i}) \\ &\quad \times \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a'_i) V^*(\mathbf{s}^+) \\ &= r(\mathbf{s}') + \gamma \sum_{\mathbf{H}'^+} \sum_{\boldsymbol{\tau}_{\setminus i}^+} \sum_{\tau_i^+} \Pr(\mathbf{H}'^+) \Pr(\boldsymbol{\tau}_{\setminus i}^+ | \boldsymbol{\tau}_{\setminus i}, \mathbf{H}_{\setminus i}, \mathbf{a}'_{\setminus i}) \\ &\quad \times \Pr(\tau_i^+ | \tau_i, \mathbf{h}'_i, a'_i) V^*(\mathbf{s}'^+) \\ &= Q(\mathbf{s}', \mathbf{a}), \end{aligned}$$

where the second equality is derived by replacing the parameter $\mathbf{H}'^+$ with $\mathbf{H}^+$, and then using $h'_{i,m} > h_{i,m}$, $a'_i \neq m$, and $\Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a'_i) V^*(\mathbf{s}^+) = \Pr(\tau_i^+ | \tau_i, \mathbf{h}'_i, a'_i) V^*(\mathbf{s}'^+)$. $\square$

B. AoI-State Threshold Property

We consider three network scenarios with different numbers of sensors and channels as below.

1) *Two-Sensor-Single-Channel Systems*: As presented in Theorem 2, we prove that the optimal policy of a two-sensor-single-channel system is an AoI-state threshold policy in Definition 2. In other words, if the optimal action at a state is to schedule sensor n , then the optimal action is still to schedule sensor n at all states that only increase sensor n 's AoI.

Theorem 2: *The optimal policy $\pi^*(\cdot)$ of the two-sensor-single-channel system has the AoI-state threshold property in Definition 2: suppose that for state $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, the optimal action is to schedule sensor i , i.e. $a_i^* = 1$, then for state $\mathbf{s}' = (\boldsymbol{\tau}'_{(i)}, \mathbf{H})$, where $\tau'_i \geq \tau_i$, the optimal action is still $a_i^* = 1$.*

Remark 1 (Analytical Challenges): Although Theorem 2 looks simple, the proof is highly nontrivial.

First, the main difficulty of the proof is that different actions cause the transition of the AoI states in multiple dimensions, resulting in the comparison of multi-dimensional value functions. However, Lemma 2 can only compare the optimal value functions with AoI state changes within one dimension. For example, given states $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, $\mathbf{s}' = (\boldsymbol{\tau}', \mathbf{H})$, and $\mathbf{s}^\circ = (\boldsymbol{\tau}^\circ, \mathbf{H})$, where $\boldsymbol{\tau} = (\tau_i, \tau_j)$, $\boldsymbol{\tau}' = (\tau'_i, \tau_j)$, $\boldsymbol{\tau}^\circ = (\tau'_i, \tau'_j)$, $\tau'_i \geq \tau_i$, and $\tau'_j \leq \tau_j$, it is easy to have $V(\mathbf{s}) \geq V(\mathbf{s}')$, but is not possible to compare $V(\mathbf{s})$ and $V(\mathbf{s}^\circ)$.

Second, one may think that the proof of Theorem 2 is intuitive as a larger AoI of sensor i results in a lower reward, and scheduling sensor i can improve the reward more efficiently than scheduling other sensors. This is a misunderstanding. Theorem 2 is equivalent to saying that the inequality $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a})$ can be derived from $Q(\mathbf{s}, \mathbf{a}^*) \geq Q(\mathbf{s}, \mathbf{a})$, where $a_i^* = 1$. We drop the constant channel state, $\mathbf{H}$, in the optimal value functions for simplicity. Based on (12), $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a})$ is equal to

$$p_{i,1}V^*(1, \tau_j + 1) + (1 - p_{i,1})V^*(\tau'_i + 1, \tau_j + 1) \geq p_{j,1}V^*(\tau'_i + 1, 1) + (1 - p_{j,1})V^*(\tau'_i + 1, \tau_j + 1), \quad (15)$$

and $Q(\mathbf{s}, \mathbf{a}^*) \geq Q(\mathbf{s}, \mathbf{a})$ is equal to

$$p_{i,1}V^*(1, \tau_j + 1) + (1 - p_{i,1})V^*(\tau_i + 1, \tau_j + 1) \geq p_{j,1}V^*(\tau_i + 1, 1) + (1 - p_{j,1})V^*(\tau_i + 1, \tau_j + 1). \quad (16)$$

We see that (15) cannot be derived directly based on (16), since it also related to the packet success rates of different sensors, i.e. $p_{i,1}$ and $p_{j,1}$.

Proof: As mentioned in Remark 1, we will prove (15) based on (16), which depends on the key technical lemma below in terms of the optimal value functions and the packet success rates, which depends on the channel states.

Definition 3 (Meet and Joint [32], [35]): Let $x' \vee x'' = \max\{x', x''\}$ and $x' \wedge x'' = \min\{x', x''\}$ denote the join and meet of two real numbers, respectively. Define $\mathbf{x}' \vee \mathbf{x}'' = (x'_1 \vee x''_1, \dots, x'_n \vee x''_n)$ and $\mathbf{x}' \wedge \mathbf{x}'' = (x'_1 \wedge x''_1, \dots, x'_n \wedge x''_n)$ as the joint and meet of the vectors $\mathbf{x}'$ and $\mathbf{x}'' \in \mathbb{R}^n$, respectively.

Lemma 3 (Probabilistic Supermodularity, $N = 2$, $M = 1$): Given states $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$ and $\mathbf{s}^\circ = (\boldsymbol{\tau}^\circ, \mathbf{H})$, where

$\boldsymbol{\tau}^\circ = (\tau''_i, \tau'_j)$, $\tau''_i \leq \tau_i$, $\tau'_j \geq \tau_j$, and $i \neq j \in \{1, 2\}$, the following holds

$$p_{j,1}V^*(\mathbf{s} \wedge \mathbf{s}^\circ) + (p_{j,1} - p_{i,1})V^*(\mathbf{s} \vee \mathbf{s}^\circ) \geq p_{j,1}V^*(\mathbf{s}) + (p_{j,1} - p_{i,1})V^*(\mathbf{s}^\circ). \quad (17)$$

Proof: See Appendix B in [34]. $\square$

In what follows, we denote the state as $\mathbf{s} = (\tau_i, \tau_j)$, since the transition of the channel states is constant during the proof. From (16), it directly follows that

$$p_{i,1}V^*(1, \tau_j + 1) \geq p_{j,1}V^*(\tau_i + 1, 1) - (p_{j,1} - p_{i,1})V^*(\tau_i + 1, \tau_j + 1). \quad (18)$$

By applying Lemma 3 to the right-hand side of (18), we have

$$p_{i,1}V^*(1, \tau_j + 1) \geq p_{j,1}V^*(\tau'_{i,1} + 1, 1) - (p_{j,1} - p_{i,1})V^*(\tau'_i + 1, \tau_j + 1),$$

which is exactly (15). Thus, we have proved $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a})$. $\square$

2) *Multi-Sensor-Single-Channel Systems*: It is intractable to prove that the optimal policy has the AoI-state threshold property in this case. Similar to the two-sensor case, we first try to prove the probabilistic supermodularity. However, there are more than two AoI states changing when applying value iterations, making it difficult to find a set of useful inequalities to prove the target inequality. We can also construct some other inequalities in terms of the optimal value function and the packet success rates that can be used to prove the threshold property, but these inequalities cannot be proved either.

Although the AoI-state threshold property of the optimal policy is not fully derived, we prove an asymptotic structural property as below.

Theorem 3: *The optimal policy $\pi^*(\cdot)$ of a multi-sensor-single-channel system with $N > 2$ has an asymptotic AoI-state threshold property: suppose that for state $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, the optimal action is to transmit the estimation of sensor i , i.e. $a_i^* = 1$, then for all states $\mathbf{s}' = (\boldsymbol{\tau}'_{(i)}, \mathbf{H})$, where $\tau'_i \gg \tau_i$, the optimal policy is still $a_i^* = 1$.*

Theorem 3 shows that the threshold structure still exists in the optimal policy of a multi-sensor-single-channel system, at least in the large AoI domain.

Proof: This theorem is equivalent to stating that if for state $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, the inequality $Q(\mathbf{s}, \mathbf{a}^*) \geq Q(\mathbf{s}, \mathbf{a})$, $\forall \mathbf{a} \in \mathcal{A}$ exists, where $a_i^* = 1$, then for state $\mathbf{s}' = (\boldsymbol{\tau}'_{(i)}, \mathbf{H})$, where $\tau'_i \gg \tau_i$, we have $Q(\mathbf{s}', \mathbf{a}^*) \geq Q(\mathbf{s}', \mathbf{a})$. Similar to Theorem 2, we need to develop the following lemmas yet in an asymptotic manner.

Lemma 4: *Given states $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$ and $\mathbf{s}' = (\boldsymbol{\tau}'_{(i)}, \mathbf{H})$, where $\tau'_i \gg \tau_i$, the following holds*

$$V^*(\mathbf{s}') \ll V^*(\mathbf{s}).$$

Proof: See Appendix C in [34]. $\square$

Based on Lemma 4, we derive the asymptotic probabilistic supermodularity below.

Lemma 5 (Asymptotic Probabilistic Supermodularity, $N > 2$, $M = 1$): *Given states $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$ and $\mathbf{s}^\circ = (\boldsymbol{\tau}^\circ, \mathbf{H})$, where $\boldsymbol{\tau} = (\tau_1, \dots, \tau_i, \dots, \tau_j, \dots, \tau_N)$ and*

$\tau^\circ = (\tau_1, \dots, \tau_i'', \dots, \tau_j', \dots, \tau_N)$ with $\tau_i \gg \tau_i''$, $\tau_j \leq \tau_j'$, and $i \neq j \in \{1, \dots, N\}$, the probabilistic supermodularity in (17) holds.

Proof: See Appendix D in [34]. $\square$

In the following, we will prove $Q(s', \mathbf{a}^*) \geq Q(s', \mathbf{a})$ based on $Q(s, \mathbf{a}^*) \geq Q(s, \mathbf{a})$ and similar to the proof of Theorem 2, we write the state as $\mathbf{s} = (\tau_i, \tau_j, \tau_{\setminus i,j})$. Using (12), $Q(s, \mathbf{a}^*) \geq Q(s, \mathbf{a})$ is equal to

$$\begin{aligned} & p_{i,1} V^*(1, \tau_j + 1, \tau_{\setminus i,j}^+) \\ & \geq p_{j,1} V^*(\tau_i + 1, 1, \tau_{\setminus i,j}^+) \\ & \quad - (p_{j,1} - p_{i,1}) V^*(\tau_i + 1, \tau_j + 1, \tau_{\setminus i,j}^+). \end{aligned} \quad (19)$$

By applying Lemma 5 to the right-hand side of (19), we have

$$\begin{aligned} & p_{i,1} V^*(1, \tau_j + 1, \tau_{\setminus i,j}^+) \\ & \geq p_{j,1} V^*(\tau_i' + 1, 1, \tau_{\setminus i,j}^+) \\ & \quad - (p_{j,1} - p_{i,1}) V^*(\tau_i' + 1, \tau_j + 1, \tau_{\setminus i,j}^+), \end{aligned}$$

and thus

$$\begin{aligned} & p_{i,1} V^*(1, \tau_j + 1, \tau_{\setminus i,j}^+) + (1 - p_{i,1}) V^*(\tau_i' + 1, \tau_j + 1, \tau_{\setminus i,j}^+) \\ & \geq p_{j,1} V^*(\tau_i' + 1, 1, \tau_{\setminus i,j}^+) \\ & \quad + (1 - p_{j,1}) V^*(\tau_i' + 1, \tau_j + 1, \tau_{\setminus i,j}^+), \end{aligned}$$

which is exactly $Q(s', \mathbf{a}^*) \geq Q(s', \mathbf{a})$ based on (12). $\square$

3) *Multi-Sensor-Multi-Channel Systems:* This case is more complex than the single-channel one, due to the increased state dimension introduced by the multi-dimensional channel. Thus, the AoI-state threshold property of the optimal policy cannot be derived strictly. In the following, we develop another structural property of the optimal policy.

Proposition 1: If for state $\mathbf{s} = (\tau, \mathbf{H})$, the optimal action is $\mathbf{a}^* = (a_1^*, \dots, a_i^*, \dots, a_N^*)$, $a_i^* = m$, and for state $\mathbf{s}' = (\tau'_{(i)}, \mathbf{H})$, where $\tau_i' \geq \tau_i$, the optimal channel assignments except for channel m keep constant, i.e. $a_n^* = a_n'^*$, $\forall a_n^* \neq 0$, and $i \neq n$, then the optimal action for sensor j at state $\mathbf{s}'$ is to not occupy channel m , if sensor j is not scheduled at state $\mathbf{s}$, i.e., $a_j^* = 0$, and its channel condition is worse than that of sensor i , i.e., $h_{i,m} \geq h_{j,m}$.

Proposition 1 says that sensor i 's allocated channel won't be occupied by other sensors with worse channel conditions when sensor i 's AoI increases and the optimal channel assignment rules for other channels remain the same.

Proof: This proposition equivalent to that if for state $\mathbf{s} = (\tau, \mathbf{H})$, the inequality $Q(s, \mathbf{a}^*) \geq Q(s, \mathbf{a})$ exists, where $a_i^* = m$ and $a_j^* = 0$, then for state $\mathbf{s}' = (\tau'_{(i)}, \mathbf{H})$, where $\tau_i' \geq \tau_i$, the following inequality holds $Q(s', \mathbf{a}^*) \geq Q(s', \mathbf{a}')$, where $a_j' = m$, $\forall h_{i,m} \geq h_{j,m}$ and $a_{\setminus i,j}^* = a_{\setminus i,j}'$.

Therefore, we will prove $Q(s', \mathbf{a}^*) \geq Q(s', \mathbf{a}')$ based on $Q(s, \mathbf{a}^*) \geq Q(s, \mathbf{a})$ in the following. Taking (14) into $Q(s, \mathbf{a}^*) \geq Q(s, \mathbf{a}')$, we have

$$\begin{aligned} & \sum_{\tau_{\setminus i,j}^+} \sum_{\mathbf{H}^+} \sum_{\tau_i^+} \sum_{\tau_j^+} \Pr(\tau_{\setminus i,j}^+ | \tau_{\setminus i,j}, \mathbf{H}_{\setminus i,j}, \mathbf{a}_{\setminus i,j}^*) \Pr(\mathbf{H}^+) \\ & \quad \times \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a_i^*) \Pr(\tau_j^+ | \tau_j, \mathbf{h}_j, a_j^*) V^*(\mathbf{s}^+) \end{aligned}$$

$$\begin{aligned} & \geq \sum_{\tau_{\setminus i,j}^+} \sum_{\mathbf{H}^+} \sum_{\tau_i^+} \sum_{\tau_j^+} \Pr(\tau_{\setminus i,j}^+ | \tau_{\setminus i,j}, \mathbf{H}_{\setminus i,j}, \mathbf{a}_{\setminus i,j}^*) \Pr(\mathbf{H}^+) \\ & \quad \times \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a_i^*) \Pr(\tau_j^+ | \tau_j, \mathbf{h}_j, a_j^*) V^*(\mathbf{s}^+). \end{aligned} \quad (20)$$

Based on $a_{\setminus i,j}^* = a_{\setminus i,j}'$, we have

$$\Pr(\tau_{\setminus i,j}^+ | \tau_{\setminus i,j}, \mathbf{H}_{\setminus i,j}, \mathbf{a}_{\setminus i,j}^*) = \Pr(\tau_{\setminus i,j}^+ | \tau_{\setminus i,j}, \mathbf{H}_{\setminus i,j}, \mathbf{a}_{\setminus i,j}'). \quad (21)$$

Thus, from (20), the following inequality can be derived

$$\begin{aligned} & \sum_{\tau_i^+} \sum_{\tau_j^+} \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a_i^*) \Pr(\tau_j^+ | \tau_j, \mathbf{h}_j, a_j^*) V^*(\mathbf{s}^+) \\ & \geq \sum_{\tau_i^+} \sum_{\tau_j^+} \Pr(\tau_i^+ | \tau_i, \mathbf{h}_i, a_i') \Pr(\tau_j^+ | \tau_j, \mathbf{h}_j, a_j') V^*(\mathbf{s}^+). \end{aligned}$$

For notation simplicity, the state can be rewritten as $\mathbf{s} = (\tau_i, \tau_j)$, because the states except for τ_i and τ_j are constant. Based on $a_i^* = m$, $a_j^* = 0$, $a_i' = 0$, and $a_j' = m$, the left-hand side and the right hand side of (20) are equal to

$$\begin{cases} p_{i,m} V^*(1, \tau_j + 1) + (1 - p_{i,m}) V^*(\tau_i + 1, \tau_j + 1) \\ p_{j,m} V^*(\tau_i + 1, 1) + (1 - p_{j,m}) V^*(\tau_i + 1, \tau_j + 1). \end{cases} \quad (22)$$

Then, the following inequality can be derived

$$\begin{aligned} & p_{i,m} V^*(1, \tau_j + 1) \\ & \geq p_{j,m} V^*(\tau_i + 1, 1) + (p_{i,m} - p_{j,m}) V^*(\tau_i + 1, \tau_j + 1) \\ & \geq p_{j,m} V^*(\tau_i' + 1, 1) + (p_{i,m} - p_{j,m}) V^*(\tau_i' + 1, \tau_j + 1), \end{aligned} \quad (23)$$

where the first inequality is derived by (20), (21), and (22), and the second inequality is by using Lemma 2 and $p_{i,m} \geq p_{j,m}$. The inequality (23) can be rewritten as

$$\begin{aligned} & p_{i,m} V^*(1, \tau_j + 1) + (1 - p_{i,m}) V^*(\tau_i' + 1, \tau_j + 1) \\ & \geq p_{j,m} V^*(\tau_i' + 1, 1) + (1 - p_{j,m}) V^*(\tau_i' + 1, \tau_j + 1). \end{aligned} \quad (24)$$

According to the analysis from (20) to (22), it directly follows $Q(s', \mathbf{a}^*) \geq Q(s', \mathbf{a}')$ from (24). $\square$

Remark 2 (Generality of the derived results): We note that all the theoretical results proved in this section merely rely on two features of the formulated optimal scheduling problem: 1) the reward function of each user is a monotonically decreasing function in terms of its AoI state, and 2) the total reward function of the problem is the sum of individual rewards. In other words, for other scheduling problems of different systems, if the two features are satisfied, the structural properties still hold. For example, if a scheduling problem is for minimizing the overall expected sum AoI (i.e., the instantaneous reward is $r(\mathbf{s}_t) = \sum_{n=1}^N -\tau_{n,t}$), not the sum estimation MSE, then the optimal policy does have the same structural properties. Similarly, considering an AoI violation minimization problem for delay-sensitive applications with an AoI threshold d [36], the instantaneous reward can be defined as $r(\mathbf{s}_t) = \sum_{n=1}^N -\mathbb{1}(\tau_{n,t} > d)$, satisfying the two features.

Remark 3 (Counterexample): We can show that if the considered optimal scheduling problem of a multi-sensor remote

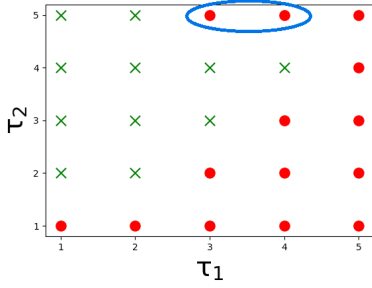


Fig. 3. The optimal policy of a two-sensor-single-channel scheduling problem with the multiplicative reward function, where $\bullet$ and $\times$ represent the schedule of sensor 1 and 2, respectively.

estimation system has a reward function as the negative product of the individual estimation MSE, i.e., $r(\mathbf{s}) = -\prod_{i=1}^N \text{Tr}(\mathbf{P}_{i,t})$, which does not satisfy the second feature in Remark 2, then the threshold structure may not exist. Fig 3 illustrates the optimal scheduling policy of a two-sensor-single-channel remote estimation system with the constructed reward function. We see that the optimal policy is not a threshold policy as in Theorem 2.

V. STRUCTURE-ENHANCED DRL

In the literature, DQN and DDPG are the most commonly adopted off-policy DRL algorithms for solving optimal scheduling problems (see [15], [19] and references therein), and they provide significant performance improvements over heuristic policies. Next, we develop threshold structure-enhanced (SE) DQN and DDPG for solving Problem 1 based on the structural properties in Definitions 1 and 2. The performance improvement of SE algorithms will be presented in Section VI.⁸

A. Structure-Enhanced DQN

Based on (11) and (12), we have the Bellman equation in terms of the Q-values of the optimal policy:

$$Q(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t) + \mathbb{E}_{\mathbf{s}_{t+1}} \left[\gamma \max_{\mathbf{a}_{t+1}} Q(\mathbf{s}_{t+1}, \mathbf{a}_{t+1}) \right]. \quad (25)$$

A well-trained DQN uses a neural network (NN) with the parameter set θ to approximate $Q(\mathbf{s}_t, \mathbf{a}_t)$ by $Q(\mathbf{s}_t, \mathbf{a}_t; \theta)$ and use it to find the optimal action, i.e., $\mathbf{a}_t^* = \arg\max_{\mathbf{a}_t \in \mathcal{A}} Q(\mathbf{s}_t, \mathbf{a}_t; \theta)$. Considering the action space defined in Section III-A, the DQN has $N!/(N-M)!$ Q-value outputs of different state-action pairs. To train the DQN, one needs to sample data (consisting of states, actions, rewards, and next states), define a loss function of θ based on the collected data, and minimize the loss function to find the optimized θ . However, the conventional DQN training method has never utilized structures of optimal policies before.

To utilize the knowledge of the threshold policy structure for enhancing the DQN training performance, we propose 1) an

⁸Note that the DRL training is conducted offline, which requires knowledge of the state-transition rules in Section III-A, building on the channel statistics. Online fine-tuning techniques for model mismatching issues during the deployment stage are beyond the scope of the work.

SE action selection method based on Definitions 1 and 2 to select reasonable actions and hence enhance the data sampling efficiency; and 2) an **SE loss function definition** to add the penalty to sampled actions that do not follow the threshold structure.

Our SE-DQN training algorithm has three stages: 1) the DQN with **loose SE action selection** stage, which only utilizes part of the structural property, 2) the DQN with **tight SE action selection** stage utilizes the full structural property, and 3) the conventional DQN stage. The first two stages use the SE action selection schemes and the SE loss function to train the DQN fast, resulting in a reasonable threshold policy, and the last stage is for further policy exploration. In what follows, we present the loose and tight SE action selection schemes and the SE loss function.

1) *Loose SE Action Selection*: We randomly select an action $\mathbf{a}^\epsilon$ from the entire action space with a probability of ϵ for action exploration; with a probability of $(1 - \epsilon)$, we generate the SE action $\hat{\mathbf{a}}$ as below. For simplicity, we drop the time index when describing action selections.

The threshold structure suggests that the actions of $\mathbf{s}$ and the state with a smaller AoI or channel state are correlated. Thus, one can infer the action based on the action at the state with a smaller channel, or AoI state, based on the channel-state and the AoI-state threshold properties in Definitions 1 and 2, respectively. We only consider the AoI-state threshold property for loose SE action selection, as it is difficult to find actions that satisfy both structural properties at the beginning of training. We will utilize both of them in the tight SE action selection stage.

Given the state $\mathbf{s} = (\boldsymbol{\tau}, \mathbf{H})$, we define the state with a smaller sensor n 's AoI as $\dot{\mathbf{s}}^{(n)} = (\dot{\boldsymbol{\tau}}^{(n)}, \mathbf{H})$, where

$$\dot{\boldsymbol{\tau}}^{(n)} = (\tau_1, \dots, \tau_n - 1, \dots, \tau_N).$$

For each n , we calculate the corresponding action based on the Q-values as

$$\dot{\mathbf{a}}^{(n)} \triangleq (\dot{a}_1^{(n)}, \dots, \dot{a}_n^{(n)}, \dots, \dot{a}_N^{(n)}) = \arg \max_{\dot{\mathbf{a}}} Q(\dot{\mathbf{s}}^{(n)}, \dot{\mathbf{a}}; \theta).$$

Recall that $\dot{a}_n^{(n)} \in \{0, 1, \dots, M\}$ is the channel index assigned to sensor n at the state $\dot{\mathbf{s}}^{(n)}$.

If $\dot{a}_n^{(n)} > 0$, then the AoI-state threshold property implies that channel $\dot{a}_n^{(n)}$ or a better channel is assigned to sensor n at the state $\mathbf{s}$. We define the set of channels with better quality as

$$\mathcal{M}^{(n)} \triangleq \{m' | h_{n,m'} > h_{n,\dot{a}_n^{(n)}}, m' = 1, \dots, M\}.$$

Then, the SE action for sensor n , say $\hat{a}_n$, is randomly chosen from the set $\mathcal{M}^{(n)}$ with probability ξ , and is equal to $\dot{a}_n^{(n)}$ with probability $1 - \xi$.

If $\dot{a}_n^{(n)} = 0$, then the AoI-state threshold property cannot help with determining the action. Then, we define the action selected by the greedy policy (i.e., the conventional DQN method) at the state $\mathbf{s}$ as

$$\tilde{\mathbf{a}} \triangleq (\tilde{a}_1, \tilde{a}_2, \dots, \tilde{a}_N) = \arg \max_{\mathbf{a}} Q(\mathbf{s}, \mathbf{a}; \theta).$$

Thus, we set the SE action of sensor n identical to the one generated by the conventional DQN method, i.e., $\hat{a}_n = \tilde{a}_n$.

Now we can define the SE action for N sensors as $\hat{\mathbf{a}} = (\hat{a}_1, \hat{a}_2, \dots, \hat{a}_N)$. If such an action meets the constraint

$$\sum_{m=1}^M \mathbb{1}(\hat{a}_n = m) \leq 1, \quad \sum_{n=1}^N \mathbb{1}(\hat{a}_n = m) \leq 1,$$

which is less restrictive than (5), we select the action $\mathbf{a}$ as $\hat{\mathbf{a}}$ and assign the unused channels randomly to unscheduled sensors; otherwise, $\mathbf{a}$ is identical to that of the conventional method as $\tilde{\mathbf{a}}$.

2) *Tight SE Action Selection*: By using the loose SE action selection, we first infer the scheduling action of sensor n at the state $\mathbf{s}$ based on the action of the state with a smaller AoI, $\hat{\mathbf{s}}^{(n)}$. Then, we check whether the loose SE action satisfies the channel-state threshold property as below.

For notation simplicity, we use $m > 0$ to denote the SE channel selection for sensor n . Given the state $\mathbf{s}$, we define the state with a smaller channel m 's state for sensor n as $\tilde{\mathbf{s}}^{(n,m)} = (\boldsymbol{\tau}, \tilde{\mathbf{H}}^{(n,m)})$, where $\tilde{\mathbf{H}}^{(n,m)}$ and $\mathbf{H}$ are identical except the element $\tilde{h}_{n,m}^{(n,m)} = h_{n,m} - 1$. Then, we calculate the corresponding action

$$\begin{aligned} \tilde{\mathbf{a}}^{(n,m)} &\triangleq (\tilde{a}_1^{(n,m)}, \dots, \tilde{a}_n^{(n,m)}, \dots, \tilde{a}_N^{(n,m)}) \\ &= \arg \max_{\tilde{\mathbf{a}}} Q(\tilde{\mathbf{s}}^{(n,m)}, \tilde{\mathbf{a}}; \boldsymbol{\theta}). \end{aligned}$$

From the channel-state and the AoI-state threshold properties, the scheduling action $\tilde{a}_n^{(n,m)}$ should be identical to m . Thus, if $\tilde{a}_n^{(n,m)} = m$, then the SE action for sensor n is $\hat{a}_n = m$; otherwise, $\hat{a}_n$ is identical to the conventional DQN action. The SE action satisfying both the threshold properties is $\hat{\mathbf{a}} = (\hat{a}_1, \hat{a}_2, \dots, \hat{a}_N)$. If such an action meets the constraint (5), then it is executed as $\mathbf{a} = \hat{\mathbf{a}}$; otherwise, we select the greedy action $\mathbf{a} = \tilde{\mathbf{a}}$.

3) *SE Loss Function*: During the training, each transition $(\mathbf{s}_t, \mathbf{a}_t, \hat{\mathbf{a}}_t, \tilde{\mathbf{a}}_t, r_t, \mathbf{s}_{t+1})$ is stored in a replay memory, where $r_t \triangleq r(\mathbf{s}_t)$ denotes the immediate reward. Different from the conventional DQN, we include both the SE action $\hat{\mathbf{a}}$ and the greedy action $\tilde{\mathbf{a}}$, in addition to the executed action $\mathbf{a}$.

Let $\mathcal{T}_i \triangleq (\mathbf{s}_i, \mathbf{a}_i, \hat{\mathbf{a}}_i, \tilde{\mathbf{a}}_i, r_i, \mathbf{s}_{i+1})$ denote the i th transition of a sampled batch from the replay memory. Same as the conventional DQN, we define the temporal-difference (TD) error of $\mathcal{T}_i$ as

$$\text{TD}_i \triangleq y_i - Q(\mathbf{s}_i, \mathbf{a}_i; \boldsymbol{\theta}), \quad (26)$$

where $y_i = r_i + \gamma \max_{\mathbf{a}_{i+1}} Q(\mathbf{s}_{i+1}, \mathbf{a}_{i+1}; \boldsymbol{\theta})$ is the estimation of Q-value at next step. This is to measure the gap between the left and right sides of the Bellman equation (25). Since the optimal scheduling policy satisfies the Bellman equation [13], i.e., $y_i - Q(\mathbf{s}_i, \mathbf{a}_i; \boldsymbol{\theta}) = 0$, the DRL algorithm has to minimize the TD error. Different from DQN, we introduce the action-difference (AD) error as below to measure the difference of Q-value between actions selected by the SE strategy and the greedy strategy:

$$\text{AD}_i \triangleq Q(\mathbf{s}_i, \hat{\mathbf{a}}_i; \boldsymbol{\theta}) - Q(\mathbf{s}_i, \tilde{\mathbf{a}}_i; \boldsymbol{\theta}). \quad (27)$$

Since the optimal policy has the threshold structure, the inferred action $\hat{\mathbf{a}}$ should be identical to the optimal action $\tilde{\mathbf{a}}$. Thus, the DRL algorithm should minimize the AD error (27).

Algorithm 1 SE-DQN for Sensor Scheduling in the Remote Estimation System

```

1: Initialize replay memory  $\mathcal{D}$  to capacity  $N$ 
2: Initialize policy network with random weights  $\boldsymbol{\theta}$ 
3: Initialize target network with weights  $\boldsymbol{\theta} = \boldsymbol{\theta}$ 
4: for episode = 1, 2, ...,  $E_1$  do
5:   Initialize state  $\mathbf{s}_0$ 
6:   for  $t = 1, 2, \dots, T$  do
7:     Generate  $\hat{\mathbf{a}}_t$  and select action  $\mathbf{a}_t$  by the loose SE action
       selection method of SE-DQN
8:     Execute action  $\mathbf{a}_t$  and observe  $r_t$  and  $\mathbf{s}_{t+1}$ 
9:     Compute action  $\tilde{\mathbf{a}}_t = \arg \max_{\tilde{\mathbf{a}}} Q(\mathbf{s}_t, \tilde{\mathbf{a}}; \boldsymbol{\theta})$ 
10:    Store transition  $(\mathbf{s}_t, \mathbf{a}_t, \hat{\mathbf{a}}_t, \tilde{\mathbf{a}}_t, r_t, \mathbf{s}_{t+1})$  in  $\mathcal{D}$ 
11:    Sample a random batch of transitions
        $(\mathbf{s}_i, \mathbf{a}_i, \hat{\mathbf{a}}_i, \tilde{\mathbf{a}}_i, r_i, \mathbf{s}_{i+1})$ 
12:    Calculate  $\text{TD}_i$  and  $\text{AD}_i$  based on (26) and (27)
13:    Update  $\boldsymbol{\theta}$  according to (29)
14:    Every  $t'$  steps set  $\boldsymbol{\theta} = \boldsymbol{\theta}_t$ 
15:   end for
16: end for
17: for episode =  $E_1, \dots, E_2$  do
18:   Repeat algorithm from line 5 to line 15 by adopting the tight
       SE action selection method
19: end for
20: for episode =  $E_2, \dots, E$  do
21:   Execute the DQN algorithm as in [15]
22: end for

```

Based on (26) and (27), we define the loss function of $\mathcal{T}_i$ as

$$L_i(\boldsymbol{\theta}) = \begin{cases} \alpha_1 \text{TD}_i^2 + (1 - \alpha_1) \text{AD}_i^2, & \text{if } \mathbf{a}_i = \hat{\mathbf{a}}_i \\ \text{TD}_i^2, & \text{otherwise,} \end{cases}$$

where α_1 is a hyperparameter to balance the importance of TD_i and AD_i . In other words, if the SE action is executed, both the TD and AD errors are taken into account; otherwise, the conventional TD-error-based loss function is adopted.

Given the batch size B , the overall loss function is

$$L(\boldsymbol{\theta}) = \frac{1}{B} \sum_{i=1}^B L_i(\boldsymbol{\theta}). \quad (28)$$

To optimize $\boldsymbol{\theta}$, we adopt the well-known gradient descent method and calculate the gradient as below

$$\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) = \frac{1}{B} \sum_{i=1}^B \nabla_{\boldsymbol{\theta}} L_i(\boldsymbol{\theta}), \quad (29)$$

where $\nabla_{\boldsymbol{\theta}} L_i(\boldsymbol{\theta})$ is given as

$$\begin{aligned} \nabla_{\boldsymbol{\theta}} L_i(\boldsymbol{\theta}) &= \begin{cases} -2((1 - \alpha_1) \text{AD}_i \nabla_{\boldsymbol{\theta}} (Q(\mathbf{s}_i, \hat{\mathbf{a}}_i; \boldsymbol{\theta}) - Q(\mathbf{s}_i, \tilde{\mathbf{a}}_i; \boldsymbol{\theta})) \\ \quad + \alpha_1 \text{TD}_i \nabla_{\boldsymbol{\theta}} Q(\mathbf{s}_i, \mathbf{a}_i; \boldsymbol{\theta})), & \text{if } \mathbf{a}_i = \hat{\mathbf{a}}_i \\ -2(\text{TD}_i \nabla_{\boldsymbol{\theta}} Q(\mathbf{s}_i, \mathbf{a}_i; \boldsymbol{\theta})), & \text{otherwise.} \end{cases} \end{aligned}$$

The details of the SE-DQN algorithm are given in Algorithm 1.

B. Structure-Enhanced DDPG

Different from the DQN, which has one NN to estimate the Q-value, a DDPG agent has two NNs [37], a critic NN with

parameter θ and an actor NN with the parameter μ . In particular, the actor NN approximates the optimal policy $\pi^*(s)$ by $\mu(s; \mu)$, while the critic NN approximates the Q-value of the optimal policy $Q(s, a)$ by $Q(s, a; \theta)$. In general, the critic NN judges whether the generated action of the actor NN is good or not, and the latter can be improved based on the former. The critic NN is updated by minimizing the TD error similar to the DQN. The actor-critic framework enables DDPG to solve MDPs with continuous and large action space, which cannot be handled by the DQN.

To solve our scheduling problem with a discrete action space, we adopt an action mapping scheme similar to the one adopted in [19]. We set the direct output of the actor NN with N continuous values, ranging from -1 to 1 , corresponding to sensors 1 to N , respectively. Recall that the DQN has $N!/(N-M)!$ outputs. The N values are sorted in descending order. The sensors with the highest M ranking are assigned to channels 1 to M , respectively. The corresponding ranking values are then linearly normalized to $[-1, 1]$ as the output of the final outputs of the actor NN, named as the virtual action $\mathbf{v}$. It directly follows that the virtual action $\mathbf{v}$ and the real scheduling action $\mathbf{a}$ can be mapped from one to the other directly. Therefore, we use the virtual action $\mathbf{v}$, instead of the real action $\mathbf{a}$, when presenting the SE-DDPG algorithm.

Similar to the SE-DQN, the SE-DDPG has the loose SE-DDPG stage, the tight SE-DDPG stage, and the conventional DDPG stage. The i th sampled transition is denoted as

$$\mathcal{T}_i \triangleq (s_i, \mathbf{v}_i, \hat{\mathbf{v}}_i, \tilde{\mathbf{v}}_i, r_i, s_{i+1}).$$

We present the SE action selection method and the SE loss function in the sequel.

1) *SE Action Selection*: The general action selection approach for the SE-DDPG is identical to that of the SE-DQN, by simply converting $\mathbf{v}_i$, $\mathbf{v}_i^\epsilon$, $\tilde{\mathbf{v}}_i$, and $\hat{\mathbf{v}}_i$ to $\mathbf{a}_i$, $\mathbf{a}_i^\epsilon$, $\tilde{\mathbf{a}}_i$, and $\hat{\mathbf{a}}_i$, respectively. Different from DQN with ϵ -greedy actions, the action generated by the DDPG based on the current state is

$$\tilde{\mathbf{v}}_i = \mu(s_i; \mu), \quad (30)$$

and the random action $\mathbf{v}_i^\epsilon$ was generated by adding noise to the original continuous output of the actor NN.

2) *SE Loss Function*: Different from the SE-DQN, the SE-DDPG needs different loss functions for updating the critic NN and the actor NN. For the critic NN, we use the same loss function as in (28), and thus the gradient for the critic NN update is identical to (29). Note that for DDPG, the next virtual action $\tilde{\mathbf{v}}_{i+1}$ is the direct output of the actor NN given the next state s_{i+1} , i.e., $\tilde{\mathbf{v}}_{i+1} = \mu(s_{i+1}; \mu)$. Thus, when calculating the TD error (26), we have $y_i = r_i + \gamma Q(s_{i+1}, \mu(s_{i+1}; \mu); \theta)$.

For the actor NN, we introduce the difference between actions selected by the SE strategy $\hat{\mathbf{v}}_i$ and the actor NN $\tilde{\mathbf{v}}_i$, when $\hat{\mathbf{v}}_i$ is executed, i.e., $\mathbf{v}_i = \hat{\mathbf{v}}_i$. If the SE action is not selected, then the loss function is the Q-value given the state-action pair, which is identical to the conventional DDPG. Given the hyperparameter α_2 , the loss function for

Algorithm 2 SE-DDPG for Sensor Scheduling in the Remote Estimation System

```

1: Initialize replay memory  $\mathcal{D}$  to capacity  $N$ 
2: Initialize actor network and critic network with random weights  $\mu$  and  $\theta$ 
3: Initialize target network and with weight  $\hat{\mu} = \mu$ ,  $\hat{\theta} = \theta$ 
4: for episode  $= 1, 2, \dots, E_1$  do
5:   Initialize a random noise  $\mathcal{N}$  for action exploration
6:   Initialize state  $s_0$ 
7:   for  $t = 1, 2, \dots, T$  do
8:     Generate  $\hat{\mathbf{v}}_t$  and select action  $\mathbf{v}_t$  by the loose SE action selection method of SE-DDPG
9:     Mapping virtual action  $\mathbf{v}_t$  to real action  $\mathbf{a}_t$ 
10:    Execute action  $\mathbf{a}_t$  and observe  $r_t$  and  $s_{t+1}$ 
11:    Compute action  $\tilde{\mathbf{v}}_t = \mu(s_t; \mu)$ 
12:    Store transition  $(s_t, \mathbf{v}_t, \hat{\mathbf{v}}_t, \tilde{\mathbf{v}}_t, r_t, s_{t+1})$  in  $\mathcal{D}$ 
13:    Sample a random batch of transitions  $(s_i, \mathbf{v}_i, \hat{\mathbf{v}}_i, \tilde{\mathbf{v}}_i, r_i, s_{i+1})$ 
14:    Calculate  $\text{TD}_i$  and  $\text{AD}_i$  based on (26) and (27) but with the virtual actions  $\mathbf{v}_i, \hat{\mathbf{v}}_i, \tilde{\mathbf{v}}_i$ 
15:    Update  $\theta$  according to (29)
16:    Update  $\mu$  according to (33)
17:    Update the target network:
        
$$\hat{\mu} \leftarrow \delta\mu + (1 - \delta)\hat{\mu}$$

        
$$\hat{\theta} \leftarrow \delta\theta + (1 - \delta)\hat{\theta}$$

18:   end for
19: end for
20: for episode  $E_1 = 1, 2, \dots, E_2$  do
21:   Repeat algorithm from line 5 to line 18 by adopting the tight SE action selection method
22: end for
23: for episode  $E_2 = 1, 2, \dots, E$  do
24:   Execute the conventional DDPG algorithm as in [37]
25: end for

```

the transition $\mathcal{T}_i$ is defined as

$$L_i(\mu) = \begin{cases} -\alpha_2 Q(s_i, \tilde{\mathbf{v}}_i; \theta) + (1 - \alpha_2) (\mathbf{v}_i - \tilde{\mathbf{v}}_i)^2, & \text{if } \mathbf{v}_i = \hat{\mathbf{v}}_i \\ -Q(s_i, \tilde{\mathbf{v}}_i; \theta), & \text{otherwise,} \end{cases} \quad (31)$$

and hence the overall loss function given the sampled batch is

$$L(\mu) = \frac{1}{B} \sum_{i=1}^B L_i(\mu). \quad (32)$$

By replacing (30) and (31) in (32) and applying the chain rule, we can derive the gradient of the overall loss function in terms of μ as

$$\nabla_{\mu} L(\mu) = \frac{1}{B} \sum_{i=1}^B \nabla_{\mu} L_i(\mu), \quad (33)$$

where $\nabla_{\mu} L_i(\mu)$ is given by:

$$\nabla_{\mu} L_i(\mu) = \begin{cases} -\alpha_2 \nabla_{\tilde{\mathbf{v}}_i} Q(s_i, \tilde{\mathbf{v}}_i; \theta) \nabla_{\mu} \mu(s_i; \mu) - 2(1 - \alpha_2) \times (\mathbf{v}_i - \mu(s_i; \mu)) \nabla_{\mu} \mu(s_i; \mu), & \text{if } \mathbf{v}_i = \hat{\mathbf{v}}_i \\ -\nabla_{\tilde{\mathbf{v}}_i} Q(s_i, \tilde{\mathbf{v}}_i; \theta) \nabla_{\mu} \mu(s_i; \mu), & \text{otherwise,} \end{cases}$$

where the action $\tilde{v}_i$ is replaced by $\mu(s_i; \mu)$ based on (30). The details of the proposed SE-DDPG algorithm are given in Algorithm 2.

VI. NUMERICAL EXPERIMENTS

In this section, we evaluate and compare the performance of the proposed SE-DQN and SE-DDPG with the benchmark DQN and DDPG.

A. Experiment Setups

Our numerical experiments run on a computing platform with an Intel Core i5 9400F CPU @ 2.9 GHz with 16GB RAM and an NVIDIA RTX 2070 GPU. For the remote estimation system, we set the dimensions of the process state and the measurement as $l_n = 2$ and $e_n = 1$, respectively. The system matrices $\{\mathbf{A}_n\}$ are randomly generated with the spectral radius within the range of $(1, 1.4)$. The entries of $\mathbf{C}_n$ are drawn uniformly from the range $(0, 1)$, $\mathbf{W}_n$ and $\mathbf{V}_n$ are identity matrices.

Since the SE-DQN and SE-DDPG require the discrete channel state space, the fading channel state is quantized into $\bar{h} = 5$ levels, and the corresponding packet drop probabilities are 0.2, 0.15, 0.1, 0.05, and 0.01. The distribution of the discrete channel states of each sensor-channel pair is generated based on Rayleigh distribution with the random scale parameter sampled uniformly from the range of $(0.5, 2)$ [38].

The NNs of the DQN and SE-DQN have the same hyperparameters, and each has two hidden layers with 256 nodes. The NNs of the DDPG and SE-DDPG are identical, and each of the actor and critic NNs has two hidden layers with 512 nodes. All the NNs in these algorithms are fully connected neural network (FCNN). The computation complexity of an FCNN is $O(\sum_{l=1}^{L-1} u_l u_{l+1})$, where u_l is the number of nodes in layer l and L is the total number of layers of the NN [39]. Based on our computing platform, the inference time of the SE-DQN and SE-DDPG are 0.02 ms and 0.04 ms, respectively. The settings of the other hyperparameters for Algorithm 1 and Algorithm 2 are summarized in Table I. Note that the input dimensions of the actor NN and the critic NN are different, as the inputs of the critic NN and the actor NN are the state vector and the state-action pair, respectively. During the training, we use the ADAM optimizer for calculating the gradient and reset the environment for each episode with $T = 500$ steps. The episode numbers for the loose SE action, the tight SE action, and the conventional DQN stages are 50, 100, and 150, respectively.

B. Performance Comparison

Fig. 4 and Fig. 5 illustrate the average sum MSE of all processes during the training achieved by the SE-DRL algorithms and the benchmarks under some system settings. Fig. 4 shows that the SE-DQN saves about 50% training episodes for the convergence, and also decreases the average sum MSE for 10% than the conventional DQN. Fig. 5 shows that the SE-DDPG saves about 30% training episodes and reduces the average sum MSE by 10%, when compared

TABLE I
SUMMARY OF HYPERPARAMETERS

Hyperparameters of SE-DQN and SE-DDPG	Value
Initial values of ϵ and ξ	1
Decay rates of ϵ and ξ	0.999
Minimum ϵ and ξ	0.01
Mini-batch size, B	128
Experience replay memory size, K	20000
Discount factor, γ	0.95
Hyperparameters of SE-DQN	
Learning rate	0.0001
Decay rate of learning rate	0.001
Target network update frequency	100
Weight of SE-DQN loss function, α_1	0.5
Input dimension of network	$N + N \times M$
Output dimension of network	$N!/(N - M)!$
Hyperparameters of SE-DDPG	
Learning rate of actor network	0.0001
Learning rate of critic network	0.001
Decay rate of learning rate	0.001
Soft parameter for target update, δ	0.005
Weight of critic network loss function, α_1	0.5
Weight of actor network loss function, α_2	0.9
Input dimension of actor network	$N + N \times M$
Output dimension of actor network	N
Input dimension of critic network	$2N + N \times M$
Output dimension of critic network	1

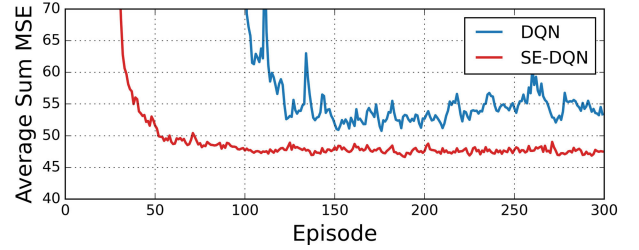


Fig. 4. Average sum MSE of all processes during training with $N = 6, M = 3$.

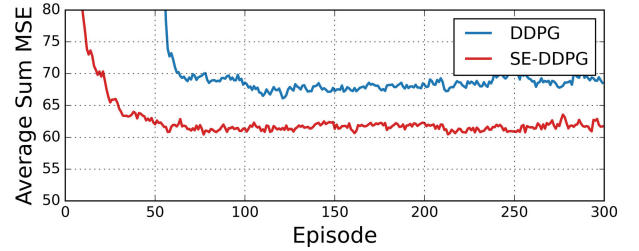


Fig. 5. Average sum MSE of all processes during training with $N = 10, M = 5$.

to the conventional DDPG. Also, we see that the conventional DQN and DDPG stages (i.e. the last 150 episodes) in Fig 4 and 5 cannot improve much of the training performance. This implies that the SE stages have converged.

In Table II, we test the performance of well-trained SE-DRL algorithms for different numbers of sensors and channels, and different settings of system parameters, i.e., $\mathbf{A}_n, \mathbf{C}_n$, and $q_1^{(n,m)}, \dots, q_{\bar{h}}^{(n,m)}$, based on 10000-step simulations. We see that for 6-sensor-3-channel systems, the SE-DQN reduces the average MSE by 10% to 25% over DQN, while both DDPG and SE-DDPG achieve similar performance as the SE-DQN. This suggests that the SE-DQN is almost optimal and that the DDPG and the SE-DDPG cannot improve further.

TABLE II
PERFORMANCE COMPARISON OF THE SE-DRL AND THE BENCHMARKS
IN TERMS OF THE AVERAGE ESTIMATION MSE

System Scale (N, M)	Param. setup	DQN	SE-DQN	DDPG	SE-DDPG
(6, 3)	1	52.4121	47.6766	48.4075	47.0594
	2	67.4247	49.8476	53.3675	44.7423
	3	84.1721	59.9127	56.9504	55.2409
	4	79.5902	65.1640	64.4313	58.5534
(6, 2)	5	55.7092	50.5983	51.3488	47.1210
	6	—	80.2522	77.4124	72.4863
	7	—	78.7715	85.3182	75.8465
	8	—	58.6024	61.4121	57.8539
(10, 5)	9	—	—	68.0247	62.4727
	10	—	—	89.6290	78.4138
	11	—	—	90.2812	81.1148
	12	—	—	—	147.2844
(20, 10)	13	—	—	181.4135	159.4321
	14	—	—	163.1257	142.1850
	15	—	—	173.0940	144.8166
	16	—	—	215.2231	160.2304

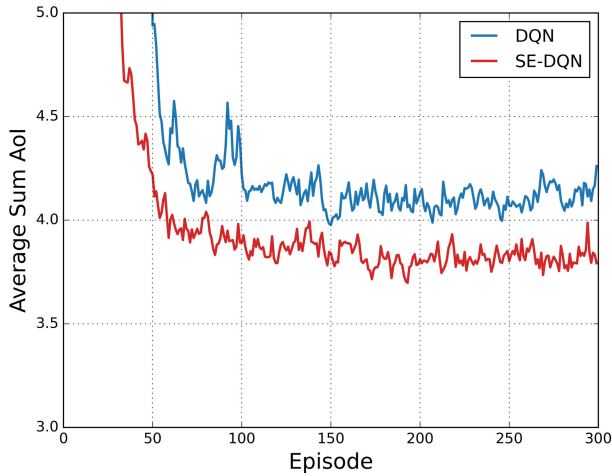


Fig. 6. Average sum AoI of all processes during the training with $N = 6, M = 3$.

In 6-sensor-2-channel systems, the DQN cannot converge in most experiments but the SE-DQN and SE-DDPG can still perform better than DDPG. In 10-sensor-5-channel systems, neither the DQN nor the SE-DQN can converge, and the SE-DDPG achieves a 10% MSE reduction over the DDPG. In particular, the SE-DDPG is the only converged algorithm in Experiment 12. In 20-sensor-10-channel systems, we see that the SE-DDPG can reduce the average MSE by 15% to 25%. Therefore, the performance improvement of the SE-DDPG appears significant for large systems.

In addition, to demonstrate that the SE-DRL algorithms can also be applied to other scenarios, we consider a multi-sensor scheduling problem for minimizing the overall expected sum AoI as mentioned in Remark 3. In Fig 6, we see that the SE-DQN can effectively reduce the average sum AoI by 10% compared with the DQN.

We also test the effectiveness of the channel-state threshold property and the loose SE action selection stage in the SE-DRL algorithms. Fig. 7 illustrates that without using the channel-threshold property in the SE-action selection process, although the convergence speed is not affected much, such a

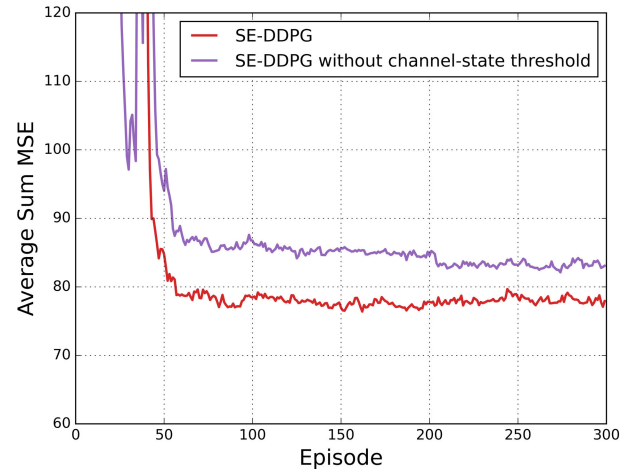


Fig. 7. Training performance comparison between SE-DDPG with and without using the channel-state threshold property, where $N = 10, M = 5$.



Fig. 8. Training performance comparison between SE-DDPG with and without the loose SE-action selection stage, where $N = 10, M = 5$.

training method does not converge to a near-optimal policy and leads to a noticeably higher estimation MSE than the original SE-DDPG. In Fig. 8, we see that if one disables the loose SE-action selection stage, the training convergence time doubles compared with the original SE-DDPG, though both converged algorithms provide similar performance. Thus, both the channel-threshold-enabled action selection and the loose action selection stage are critical to guarantee the performance of the SE-DRL algorithms.

VII. CONCLUSION

In this paper, we have proved that the optimal scheduling policy of the remote estimation system satisfies a number of threshold properties. Then, based on these theoretical guidelines, we have developed the structure-enhanced deep reinforcement learning (SE-DRL) algorithms for solving the optimal scheduling problem. In particular, we have proposed a novel action selection method and a new loss function to improve training efficiency. Our numerical results have illustrated that SE-DRL algorithms can save training time by 50% and reduce the estimation mean-square error (MSE) by 10% to 25%, when compared with benchmark DRL

algorithms. In addition, the results also show that SE-DRL can effectively solve a range of optimal scheduling problems, not limited to remote state estimation systems. For future work, we will investigate other structural properties of optimal scheduling and resource allocation policies for wireless communications and networked systems, and then use these properties to develop tailored DRL algorithms. We will develop structure-enhanced algorithms for NOMA-based sensor scheduling and resource allocation problems.

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